

Assignment 5: Reinforcement Learning

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Question 1

The suggested reward function is not a very good choice for training the robot. In fact, it gives the robot a +1 reward signal for each time step taken in the maze. If in each experiment we iterate over a number of time steps $\gg 1000$ and the robot hits the wall staying in its position and orientation, then it would achieve a total reward that is higher than the one it would get if it escaped successfully from the maze.

Question 2

It would be beneficial to provide it a +1 reward not for each time step but for each step it takes forward/backward. If it hits the wall and stops moving, then we should assign it a -1000 reward.

Moreover, we could provide it more frequent rewards so as to give it more feedback about the quality of its behaviour. For example, we could give it rewards for achieving intermediate goals, such as if it is able to take n steps without hitting the wall, with $n = 10, 20, 30, \dots$

We also need to take into account exploration and exploitation: we want our robot to explore the maze in order to find the best path to the exit, while exploiting the best options it has found so far. For this reason we could give it a small reward when it chooses a path it has not already explored.

Question 3

$$G_{t+1} = R_{t+2} + \gamma R_{t+3} + \gamma^2 R_{t+4} + \dots$$

$$\begin{aligned} G_t &= R_{t+1} + \gamma (R_{t+2} + \gamma R_{t+3} + \dots) \\ &= R_{t+1} + \gamma G_{t+1} \end{aligned}$$

Question 4

Let's consider the following sum:

$$\tilde{G} = r_{\max} + \gamma r_{\max} + \gamma^2 r_{\max} + \dots = r_{\max} \sum_{k=0}^{\infty} \gamma^k$$

Since $\tilde{G} > G_t \forall t$ and $\tilde{G}$ is a geometric series, if $|\gamma| < 1$, then

$$\tilde{G} = \frac{r_{\max}}{1 - \gamma} > G_t \quad \forall t$$

Question 5

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gamma = 0.9
R1 = -1
R2 = -0.5
R3 = 2.5
R4 = 1
R5 = 3

G5 = 0
G4 = R5 + G5
G3 = R4 + gamma * G4
G2 = R3 + gamma * G3
G1 = R2 + gamma * G2
G0 = R1 + gamma * G1

----- Result:
G0 = 3.2723000000000004,    G1 = 4.747,    G2 = 5.83,    G3 = 3.7,
G4 = 3,    G5 = 0
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Question 6

If we add a constant c to each reward G_t becomes:

$$\begin{aligned} G_t &= (R_{t+1} + c) + \gamma(R_{t+2} + c) + \gamma^2(R_{t+3} + c) + \dots \\ &= c \sum_{k=0}^{\infty} \gamma^k + R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots \end{aligned}$$

Each R_t term is shifted by a constant term c . This shift does not affect the relative importance of different terms or the overall structure of the discounted return. The discount factor γ still governs the weighting of future rewards.

Bonus Question

$$G_t = 1 + \gamma + \gamma^2 + \dots = \sum_{k=0}^{\infty} \gamma^k = \frac{1}{1 - \gamma} \quad \text{if } |\gamma| < 1$$